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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Sticks and Stones
Selected Domain	Domain 2
Selected SDG Track (1–17)	Track 8
Project / Solution Name	Resolve
Name (Leader)	Ee Jun Hao
Email & Phone (Leader)	ejhchesteree7@gmail.com (01116151223)
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's roughly 1.3 million micro, small and medium enterprises (MSMEs) make up 96% of all businesses, 39.7% of GDP and 48.7% of employment (DOSM, 2025). Most run on WhatsApp, spreadsheets and the owner's own evenings.

One daily burden is reviews and complaints. When a café customer posts "Tunggu 40 minit, staff rude gila", the owner may see it days later, argue about it in a group chat, then reply inconsistently or never. Nobody knows who may apologies or offer a voucher and repeat problems like slow Friday-night service go unnoticed.

This costs real business. In Bright Local's 2026 survey of US consumers, 80% were more likely to use a business that replies to all reviews, and 50% were put off by templated replies. Owners lose customers and time that could go to growth, the productivity gap targeted by SDG 8.2 and 8.3. Moreover, small businesses cannot afford reputation-management teams or complex software, nor let untrained staff hand out compensation freely. They need a tool that is easy to learn, keeps the owner in control and records every decision. In addition,our ideal customer profile is owners, managers and crew of Malaysian cafés and retail shops first, then salons and clinics. Their own customers also benefit, as they deserve a timely and respectful reply

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Resolve is a web application that takes one café review from "someone said something" to "fixed and logged" in six steps.

(1) Capture: a guest taps an NFC card or scans a QR code that opens a simulated review page standing in for Google.

(2) Understand: an optional GenAI model reads the review in Malay, English or Manglish and returns strict JSON (sentiment, category, severity, reason and a same-language draft reply). If the AI is slow or unavailable, a keyword fallback ("Basic mode") takes over. High-severity cases such as food safety always go to the Owner.

(3) Route: plain rules, not AI, send the case to Staff, Manager or Owner and enforce voucher limits (Manager up to RM 20, Owner above).

(4) Approve: the right person approves, edits or rejects. Nothing is posted or offered without human approval.

(5) Reply and recover: the reply appears on the review page. For unhappy guests, a Manager or Owner can send a private goodwill message with an optional voucher code through a simulated WhatsApp follow-up. Nothing leaves the app; a real version would use the WhatsApp Business Platform with guest opt-in.

(6) Record and learn: an append-only audit log records who did what and when, and a recurring-issue panel shows patterns such as "Slow service on Friday nights", calculated from categorised cases.

AI is used only to understand messy multilingual text and draft replies; rules decide who may approve what.

From the SDC workshops we apply GenAI APIs (structured JSON, prompt rules, validation), a relational Postgres schema with roles, cases and an append-only log, and cloud deployment (Vercel, Supabase) with server-side permission checks. Settings are data per tenant and outlet, so the same engine can later serve other small service businesses through Orbyte.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Yes, partly. Replit (sponsor-provided USD 20 credits) was used to build the app with Replit Agent and to host a backup deployment. It supportes development only, and the prototype does not depend on it. The live demo runs on Vercel and Supabase, which are our own free-tier accounts. No sponsor GenAI API is used: the AI step is an optional OpenAI-compatible model, switched off in the demo, with a built-in keyword fallback.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We did not interview owners. We built from published evidence on Malaysian MSMEs (DOSM, 2025) and review behaviour (BrightLocal, 2026; 1,002 US adults), and from complaint and compliment patterns in 52 synthetic reviews (21 complaints, 23 compliments, 8 neutral). Owner interviews are our first step in Phase 2.

Define: one customer, owner-run cafés (one to three outlets).

Ideate: we rejected an auto-replying chatbot because owners must stay in control. AI drafts, plain rules route cases and limit vouchers, and a person approves.

Prototype: an app on Vercel and Supabase with an NFC/QR review page, role-based approval, a simulated WhatsApp follow-up, an append-only log and a keyword fallback.

Test: automated tests cover permissions, routing and fallback. In a timed test of 10 reviews across two outlets, manual replies took 12.08 minutes in total (72.5 seconds each) versus 4.5 minutes (27 seconds each) with Resolve's draft plus human approval: 62.7% faster.

Deploy: the live site scored 125/100 (A+) on the MDN HTTP Observatory. In Phase 2 we replace the simulated sign-in, Google page and WhatsApp with Orbyte login and real integrations, then pilot with real cafés.

Sources: DOSM, MSME Performance 2025; BrightLocal Consumer Review Survey 2026.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	- Target 8.3 " foster creativity and innovation and growth of micro and small medium enterprises" - Target 8.2 " Achieve higher economic productivity through technological upgrading"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Handling one customer review by hand (reading it, writing a reply, deciding on compensation and recording it) takes about **72 seconds** of owner or manager time. This was measured by our team across 10 synthetic reviews, which took **12.08 minutes in total**. In Malaysia, micro, small and medium enterprises (MSMEs) make up about 96% of all businesses and employ 48.7% of the country's workforce (DOSM, 2025). However, most still handle customer feedback manually,through social media and review pages.

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Time taken to handle one review by replying manually against using Resolve's draft plus human approval. ~~Supporting KPIs: the percentage of cases routed to the correct approver, and the percentage of replies and vouchers with a recorded human approval.~~. Supporting KPIs are the percentage of cases routed to the correct approver and the percentage of replies and vouchers with a recorded human approval.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 60% less handling time per review (from 72.5 seconds to under 30 seconds; our 10-review test measured 62.7%). Furthermore,at least 90% correct routing on our test set; 100% of posted replies and vouchers with a logged human approval; and at least 80% of first-time users resolving a complaint unaided. The time result comes from our own test on synthetic data; the other figures are targets.

Sources:

1) Bernama. (2026, July 31). *Govt shifts focus to scaling up MSMEs after resilient 2025* [News article]. https://www.bernama.com/en/news.php?id=2588561
2) BrightLocal. (2026, February 11). *Local consumer review survey 2026*. https://www.brightlocal.com/research/consumer-review-survey/
3) Human Resources Online. (n.d.). *Malaysia's MSMEs outgrew the national economy in 2025, accounting for nearly 40% of GDP*. https://www.humanresourcesonline.net/malaysia-s-msmes-outgrew-the-national-economy-in-2025-accounting-for-nearly-40-of-gdp
4) United Nations. (n.d.). *Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all*. https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19.1	**code**	**MIT**
Vite 7	**code**	**MIT**
Tailwind CSS 4	**code**	**MIT**
Express S	**code**	**MIT**
Drizzle ORM 0.45	**code**	**Apache-2.0**
Zod 3	**code**	**MIT**
node-postgres (pg) 8	**code**	**MIT**
TanStack React Query 5	**code**	**MIT**
Radix UI	**code**	**MIT**
Lucide React	**code**	**ISC**
OpenAI SDK 7	**code**	**Apache-2.0**
gpt-4.1-nano(optional,off in live demo)	**Pre-trained model**	**OpenAI API terms**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]: 

[Date]: 23/10/2026